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Hover-Propeller Full Design Cycle - Performance and Acoustic

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This paper describes the full design cycle of designing, fabricating, and testing of hover propellers. The design case is specified using quantitative and measured goals, thus establishing the design objectives and constraints. The design process uses numeric optimization together with various analytic tools of aerodynamic and acoustic. From the analytic design, various candidates are chosen to be fabricated. These propellers are manufactured and tested on specialized rigs for their performance and acoustic characteristics. Comparing these test results validates the design procedure and shed light on the importance of propeller performance versus acoustic characteristics. Hence shows the tradeoff between these two disciplines. The comprehensive development process together with discussion on each of the design and test phases, makes the current paper a rich source for both propeller designers and hover-vehicle developers.

I. Nomenclature

A	=	disk area
c	=	chord
FM	=	figure-of-merit
I_0	=	no-load current
I_{ESC}	=	ESC output current
K_v	=	motor speed constant
P_{shaft}	=	shaft power
Q	=	torque
t	=	thickness
r	=	radial coordinate
R	=	radius, distance
R_a	=	armature resistance
V_{ESC}	=	ESC output voltage
η_e	=	electric system efficiency
η_{ESC}	=	ESC efficiency
μ	=	average
θ	=	observer azimuth angle
σ	=	standard deviation
ρ	=	air density

Acronym

AC	=	alternating current
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BEM	=	blade-element model
BPF	=	blade-pass-frequency
ESC	=	electronic speed controller
MDO	=	multidisciplinary design optimization
OASPL	=	overall sound pressure level
PM	=	permanent magnet
SPL	=	sound pressure level
UAM	=	urban air mobility

II. Introduction

Urban Air Mobility (UAM) development has been expanding since the publication of UBER-Elevate white paper published in 2016.^[1] Since then numerous manufacturers have been developing various UAM configurations for example Refs. [2] and [3] show possible configurations for such vehicles; most of which are multi-propeller based. Figure 1 shows a comparison between two of the leading UAM vehicles; on the left the EHang 216 and on the right VoloCity.

In its basis, most UAM's are eVTOL (electric vertical takeoff and landing) vehicles. The main source for thrust, especially at hover, comes from rotating-wing such as propellers, rotors, fan etc. During hover, the thrust requirement reaches its peak, i.e. the thrust equal to the vehicle weight. According to simple momentum theory,^[4] this justified high total disk-area, which improves hover capabilities, mainly decreases the required power to hover.

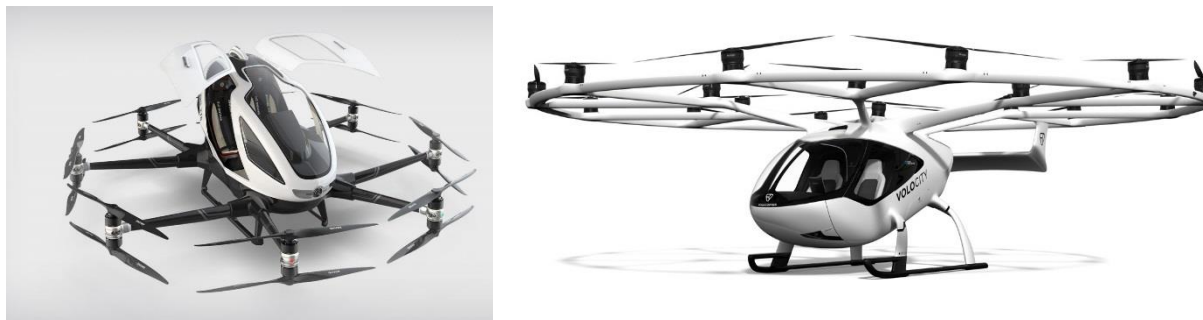


Figure 1, Comparison of EHang and VoloCity (images are scaled equally)
Left: EHang 216^[5], Right: VoloCity^[6]

This makes the propellers a critical item in these vehicles, especially at hover conditions. At hover the propulsion system performance is at its highest required power,^[7] thus propeller required power at hover impacts the overall vehicle performance. In addition, the acoustic signature at hover is the highest and together with new regulations^[8] the importance of optimized hovering propellers increase dramatically.

In the current effort, design case will be a downscale of UAM propeller. Still, all related features for the design process are treated in the same manner as for large diameter propellers. Special attention should be given to the acoustic signature which in small propeller might be ruled by the broad-band noise. In the current case only tonal noise is taken into consideration, hence broad-band noise is not considered.

III. Scope

In this paper the design and testing procedures for hover propellers is depicted. First the design procedure which includes both analyses validation and optimization will be reviewed. From this design stage, several propellers were chosen and fabricated. These propellers were then tested on specialized test rig. The tests include both performance and acoustic measurements. Finally, test results analysis was conducted and comprehensive comparison of test results to analysis validated the design procedure.

Although UAM requires high thrust, an equivalent small propeller is specified, thus the entire design, fabrication and testing procedures are simpler and more rapid. Still, all results are highly related to all hovering configuration, with the appropriate scaling.

IV. Design Specification

As a reference propeller, the Mejzlik 18×6 is used. Fig. 1 shows the Mejzlik 18×6 propeller and Fig. 2 shows its geometric properties as function of radial coordinate, r , i.e. pitch, β , chord-to-radius ratio, c/R , and thickness-ratio, t/c , distribution. According to design criteria, the propeller radius is limited to $R \leq 0.23\text{m}$, which is the radius of Mejzlik 18×6 propeller.

The propeller in this effort is specified according to its produced thrust. At design conditions, the Mejzlik 18×6 gives thrust of $T = 2.8 \text{ kgf}$ which is established as the required thrust for hover (static operation) for all future designs. The propulsion system is based on Sobek 20-38 Spider AC-motor with Kontrol-X 55LV electronic speed controller, ESC. The acoustic signature is optimized for an observer which is located at azimuth angle, $\theta=100^\circ$, relative to the propeller axis, as depicted in Fig. 3. The distance between propeller and observer is not a consideration. Unless atmospheric attenuation is considered, acoustic improvements between different propellers, are not depend on the propeller-observer distance.

The above specifications allow the design of propeller with various goals. The most important is the required battery power, electric power, P_e . Different from other design efforts, here the battery power is the most important, thus the electric propulsion system is to be considered through the design iterations.^[9] The second parameter to minimized is the acoustic signature as heard by the observer. The tradeoff between these two goals is to be found using optimization.

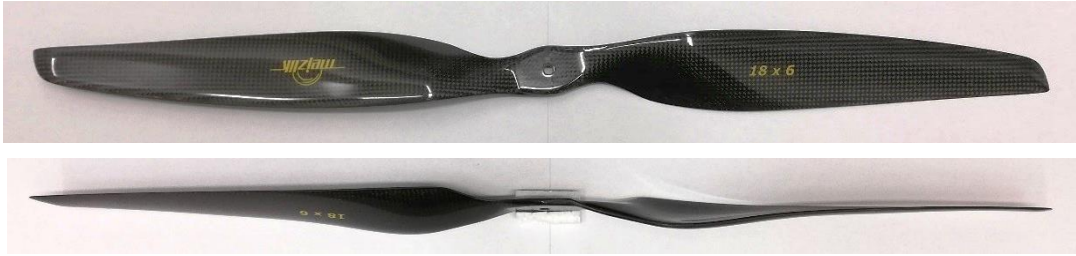


Fig. 1 Mejzlik 18×6 Propeller, front and top views

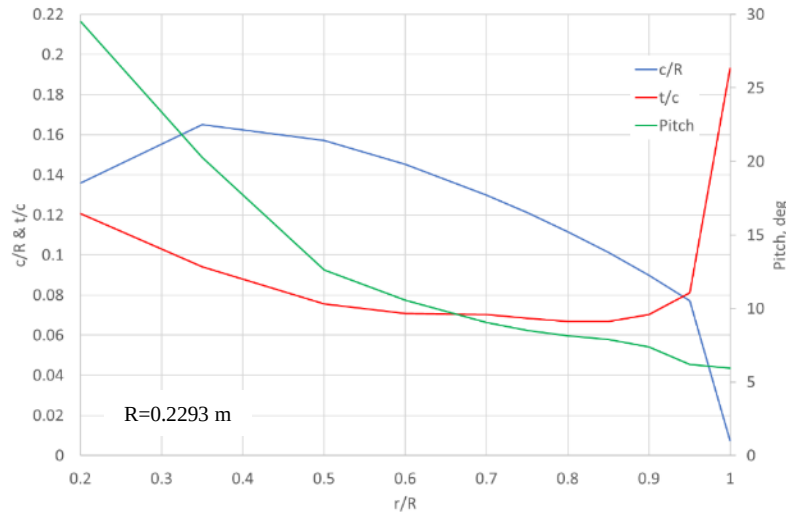


Fig. 2 Mejzlik 18×6 Propeller, front and side views

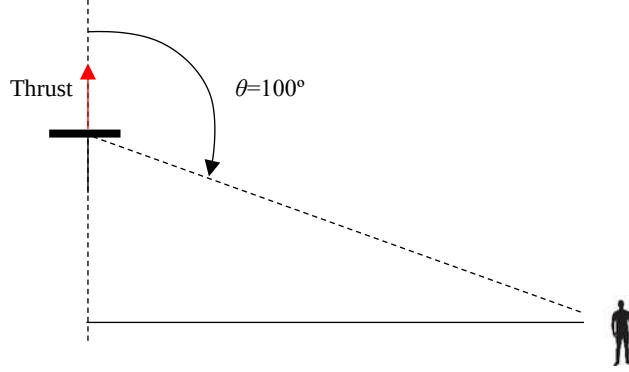


Fig. 3 design conditions of observer-propeller attitude

V. Analysis Validation

To allow proper optimization, the required analyses are to be validated. In this case, three models were validated extensively: the propeller performance model, electric system model, and propeller acoustic model.

The propeller performance model is based on blade-element model (BEM) which was extensively validated in the past.^[10] Still, in the current case, the small dimensions and low Reynolds number require revised validation. Moreover, most past validation cases were of axial flight regime, while in the current case hover condition is treated. Although hover rotor cases were validated in the past,^[9] new revised validation is required.

The performance test was conducted with 2 and 4 blades Mejzlik 18×6 propeller, i.e. both 2 & 4 blades share the same blade geometry. The test was conducted on a test rig which measures the thrust, T , shaft power, P_{shaft} , speed, Ω , and electric power, P_e . The electric power is based on the measurement of the input-ESC voltage and current. The validation test rig is depicted in Fig. 4. The results are presented in Fig. 5 in both shaft power and figure-of-merit, FM , versus thrust. Figure-of-merit is defined in Eq. (1).

$$FM = \frac{T^{1.5}}{P_{shaft} \sqrt{2 \cdot \rho \cdot A}} \quad (1)$$

where ρ is the air density and A is the propeller disk area.

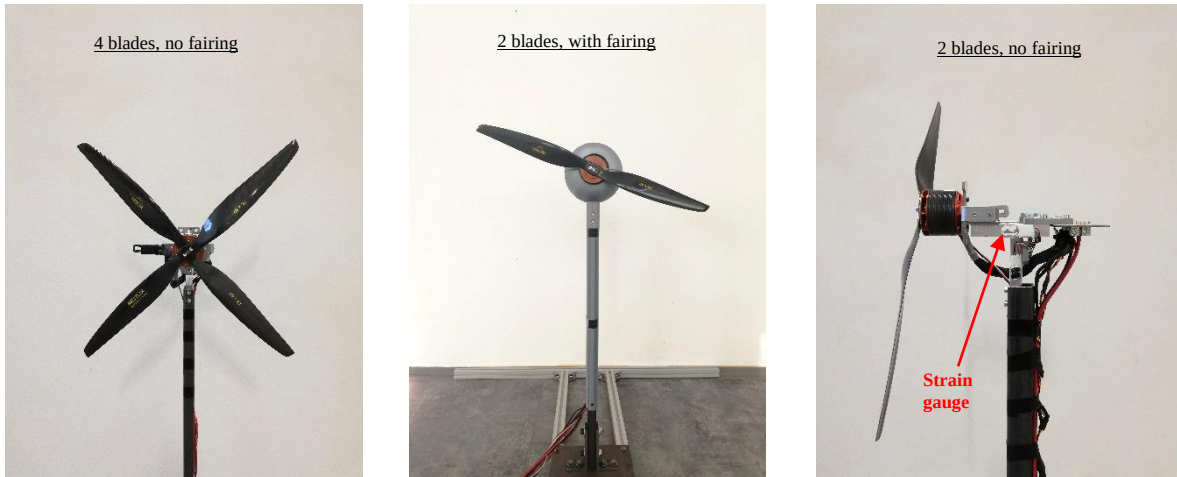


Fig. 4, 2 & 4 blades Mejzlik 18×6 propeller performance validation test rig, no fairing installed

BEM analysis uses a 2-D aerodynamic database based on the geometry of the propeller cross sectional airfoils. Accuracy of the 2-D aerodynamic database is an important part of BEM level-of-confidence. Thus, substantiation of

the current database was conducted using EZair RANS (Reynolds Average Navier-Stokes) software.^[11] A full factorial database was calculated for wide range of angle-of-attacks (-5° - 25°), Mach numbers (0-0.9), and Reynolds numbers (20,000-1,000,000). This covers all operational conditions for the blade's cross-sections. The computational domain spread for 50 chords-length and the turbulent model was 2nd order accurate Spalart-Allmaras.

According to Fig. 5, the analysis exhibits good agreement to test results. Moreover, the various trends concerning dependency of power and FM with thrust, are replicated accurately. Also the number-of-blades trend, 2 and 4 blades, is predicted accurately. This enables the usage of the BEM analysis under optimization design procedure.

In the current effort a simple motor model is used to find the required electric power. The model is based on four constant parameters: speed constant, K_v , armature resistance, R_a , no-load current, I_0 , and controller efficiency, η_c .^[12] The model is based on the following assumptions:

- Power factor is equal to unit. This assumption is applicable to small brushless Permanent Magnet (PM) motors.
- Magnetic losses (eddy/Foucault Current and magnetic hysteresis) can be neglected.

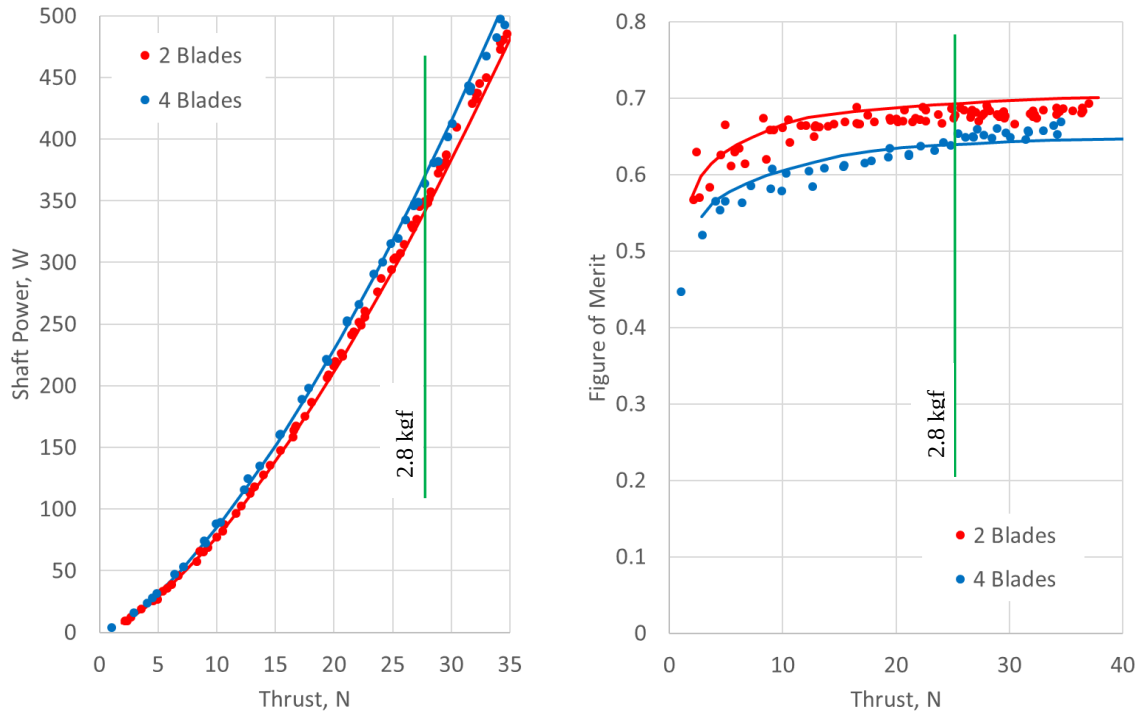


Fig. 5 2 & 4 blades Mejlík 18×6 propeller performance validation
Dots – experiment, Line - Analysis

ESC output current, I_{ESC} , and voltage, V_{ESC} , are estimated using Eq. (2).

$$I_{ESC} = \frac{P_{shaft} \cdot K_v}{\Omega} + I_0$$

$$V_{ESC} = \frac{\Omega}{K_v} + I_{in} \cdot R_a$$
(2)

Total electric power, P_e , used from the energy storage (batteries), is calculated using the ESC efficiency, η_{ESC} , as depicted in Eq. (3).

$$P_e = \frac{I_{ESC} \cdot V_{ESC}}{\eta_{ESC}}$$
(3)

Thus, the electric system efficiency, η_e , can be calculated using the ratio of P_{shaft} and P_e

$$\eta_e = \frac{P_{shaft}}{P_e}$$
(4)

The motor manufacturer suggested some values for the above parameters (K_v , R_a , I_0 , and η_{ESC}). Still, an optimized estimation was done to find the parameters values which give best agreement to the performance test. To increase the level of accuracy, the speed constant, K_v , is assumed to be linear dependent with the torque, Q , of the motor.

Comparison of test and analysis results is presented in Fig. 6. The original manufacturer parameters resulted with average η_e error of $\mu=2.4\%$ with standard deviation of $\sigma=1.8\%$. Note that for μ and σ , only the speed above 3,500 rpm are used. Below this speed, the system is far from its design conditions (of $T=2.8$ kgf).

The optimized parameters resulted with $\mu=0.75\%$ and $\sigma=0.5\%$. This agreement may be related to the optimized parameters estimation. Still, after testing other various propeller configurations (the designed propeller which are to be presented in what follows) the agreement remained good as presented in Fig. 7 ($\mu=1.0\%$ and $\sigma=0.7\%$). This validated the electric model which is used in the design process.

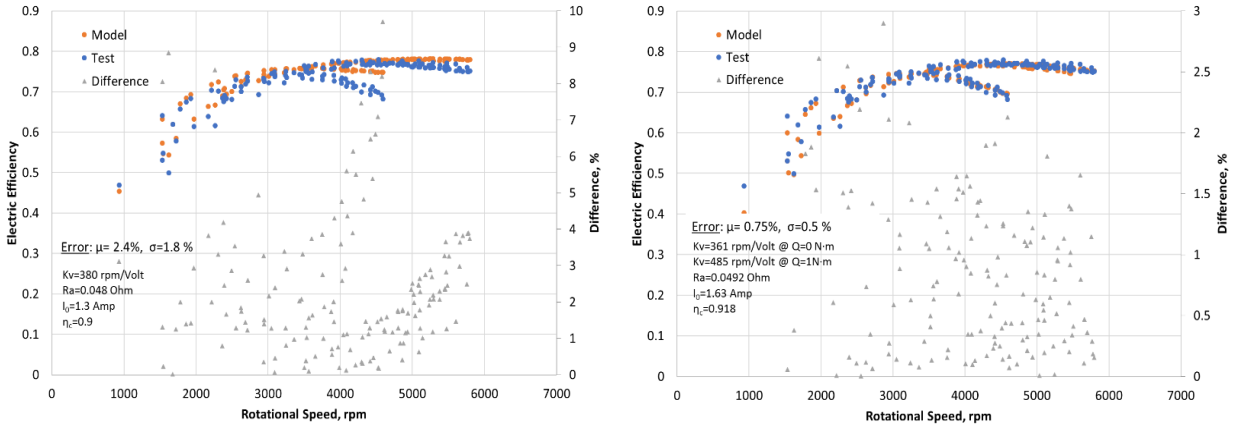


Fig. 6 electric model efficiency – test to model comparison
Left – original manufacturer parameters, Right - optimized parameters

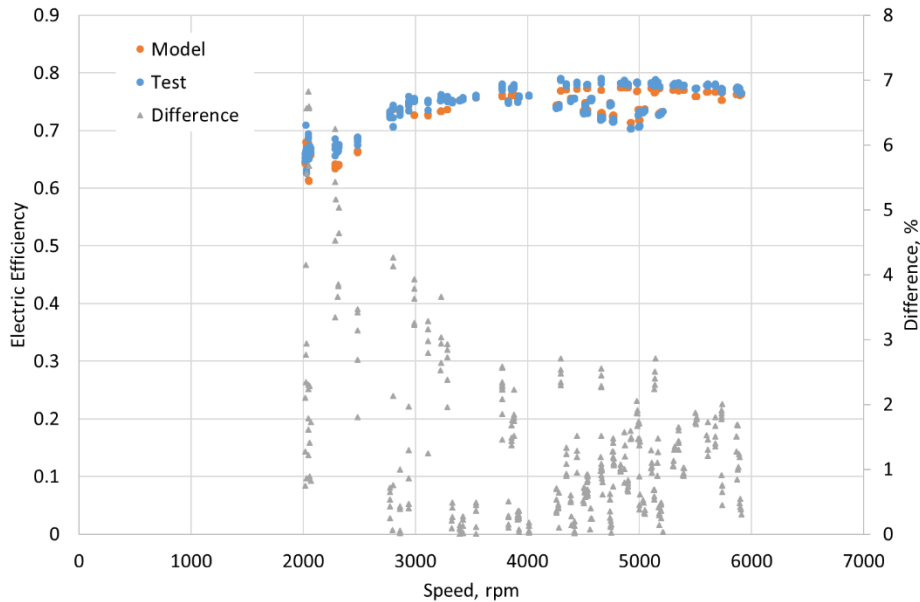


Fig. 7 electric model efficiency – results of various propellers which were not used to estimate the model

The last analysis to be validated is the acoustic analysis. The test was held in Resideo 5×5×5 m anechoic chamber, located in Brno, Czech Republic. Fig. 8 shows the faired test rig with the Mejzlik 18×6 propeller installed in the chamber. A total of 20 microphones on a standard hemisphere^[17] was hung above the propeller. The propeller was located off-center from the array and at a 45° angle so as to minimize the number of microphones which would be located in the wake or recirculation flows. As a result, a range of radii from 1.1-2.9 meters and propeller azimuthal angles 75.2-154.7° was recorded.

The current acoustic model predicts only the tonal noise of the propeller. The model is based upon Farassat formulation^[13] as used in former design cases^[14]. The model went through extensive validation for various cases of propeller on various flight regimes.^{[15], [16]}

The results for 3 of the microphones, allocated at different azimuth angles, θ , are presented on Fig. 9. The results from the test are tonal-spectra integrated from narrow-band autospectral densities ($\Delta f = 5.9$ Hz) about each of the tonal peaks (this method varies slightly from method of Farassat which uses a Δf equal to the BPF). Note that the microphones are located at different distances, R , from the propeller, thus the comparison between microphones is useless without a distance correction. Both test and analysis were conducted at the same propeller speed of $\Omega=4,980$ rpm, thus both has the same BPF (blade passing frequency). To match the same thrust (2.8 kgf) as measured in the acoustic test, the analysis was conducted with pitch change of $\Delta\beta=+0.9^\circ$, this discrepancy is small. The reason for this "thrust fitting" is the need for isolation acoustic validation with minimizing the inaccuracies of performance model. Main tonal-noise source is the thrust, thus to compare acoustic model vs. test, matching of thrust and rotational speed is essential.

The comparison of the separate harmonics between test and analysis in Fig. 9 indicates that the first, second, and sometimes the third harmonics, are predicted quite accurately. This includes the trend of decreased sound pressure level, SPL, with increased harmonic. As analysis SPL decreases below 50÷55 dB, the test's SPL remain constant at about 50÷55 dB. This behavior is not typical for propellers in a free field and is related to the test environment and test data reduction. Similar results were observed in the work of Block.^[18]

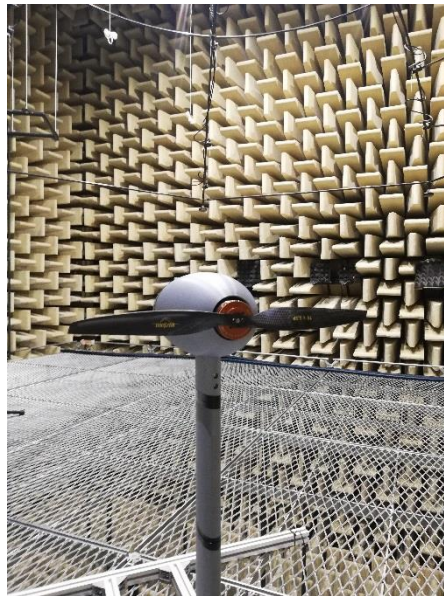


Fig. 8 Mejzlik 18×6 propeller installed on a faired rig in Resideo anechoic chamber

Some enforcement to this hypothesis is found in Ref. [18], which encounter the same type of results and stated:

Typically, the results from Method 1 and Method 2 agree for the first few harmonics, which are above the background noise of the tunnel. As the harmonic number increases, Method 2 gives the levels of the background noise, whereas Method 1 gives levels below the background noise which appear to follow the trends expected for propeller noise.

In the above quote “method 1” refers to normal behavior of propeller SPL harmonics, while “method 2” refers to results similar to what presented here. A typical example from the same reference can be found in Fig. 10 which shows test results from Ref. [18]. This exhibits the typical behavior of “method 1” versus “method 2”. Similar methods with

the same phenomena are discussed in Ref. [19]. There, several post-processing techniques are implemented to better recognize and separate the tonal and broadband noise.

A recent work,^[20] has shown these increased peaks to be resultant from unsteadiness in the flow caused by recirculation in a closed environment. In Ref. [20] the authors show the importance of air recirculation in the test chamber which can cause some of the high frequency deviation. In the current effort, the results are not treated with non-circulatory techniques, thus the higher harmonics of the test results are heavily influenced. Ref. [20] was published after the collection of the present paper's data set and hence the knowledge gained from their methods could not be applied to the current data set. Future works hope to incorporate methods influenced by these results.

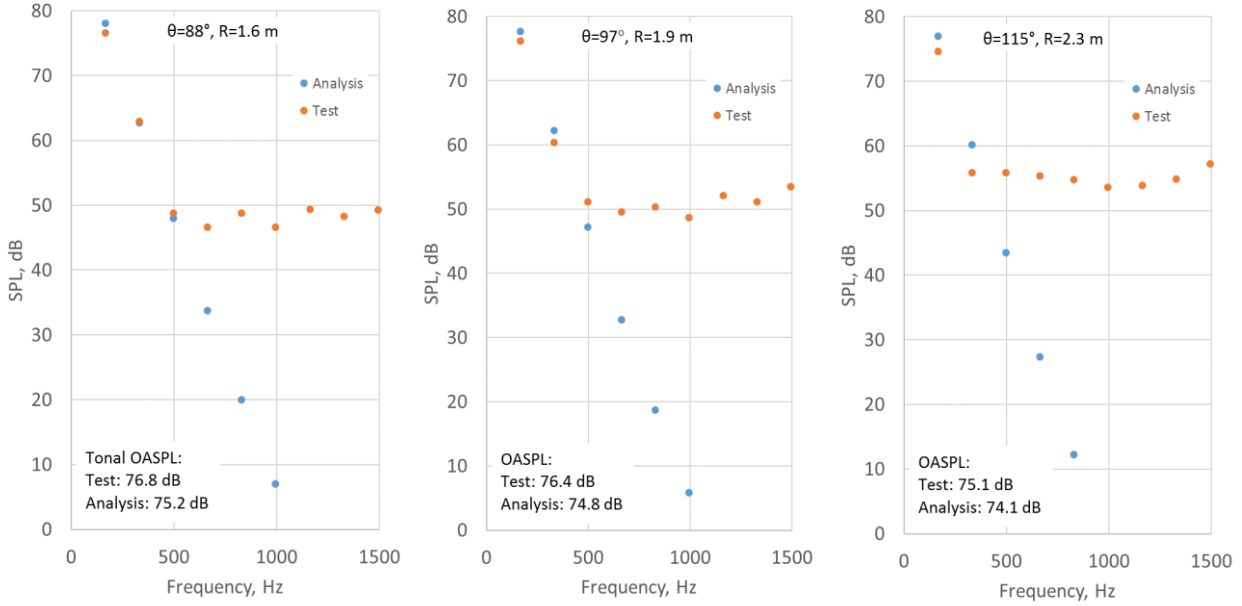


Fig. 9 Mejzlik 18x6 propeller acoustic results as function of azimuth angle

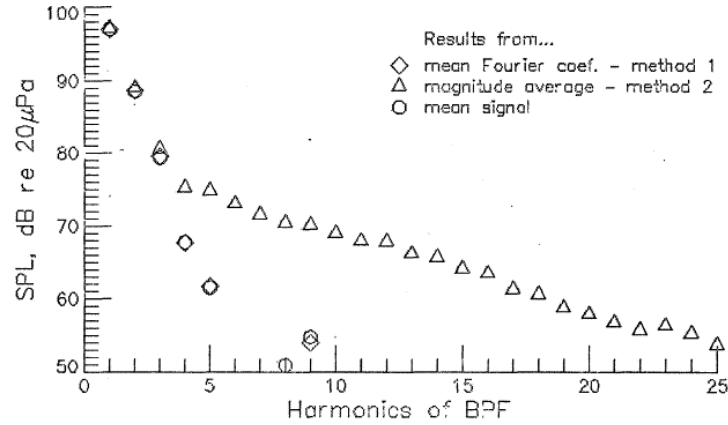


Fig. 10 Acoustic results according to “method 1” and “method 2” from Ref. [18]

Fig. 11 shows the 1st and 2nd harmonics, and the tonal overall SPL, OASPL as function of θ angles. In this chart the microphone distances are normalized to $R=1.5$ m. At low θ angles, in front and next to the propeller disk, up to about $\theta=120^\circ$, the agreement is good. At rear azimuth, $\theta=150^\circ$, the agreement deteriorates. At these rear position, the microphones were located far from the propeller, thus their 2nd harmonic and above SPLs, were about 55 db, which caused the background noise to dominate the measurements.

At the design conditions (see Fig. 3) namely $\theta=110^\circ$, the comparison shows good agreement, thus the model is validated for this design procedure.

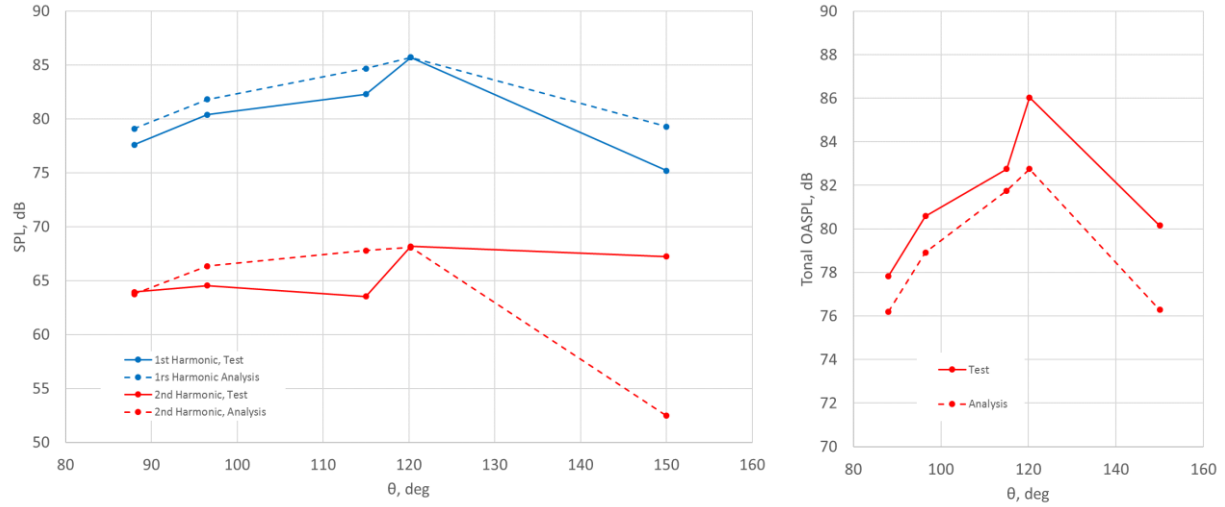


Fig. 11 Mejzlik 18×6 propeller acoustic results as function of azimuth angle, $R=1.5$ m
Left – 1st and 2nd harmonics, Right - OASPL

VI. Optimized Designs

Design technique is similar to former cases accomplished with the same tools. These tools include using the validated analysis tools (BEM, electric model, and acoustic model) together with Esteco modeFRONTIER framework.^{[15],[9]}

Fig. 12 presents a screen capture of the modeFRONTIER framework. modeFRONTIER design environment enables the integration of different simulation models into a single synergetic design tool. In addition, it allows the use of various optimization procedures, thus a MDO tool is obtained. In the current case, first the propeller performance is calculated and then the propeller acoustic is estimated. The use of modeFRONTIER enables an easy usage of any of the input or output parameters, either as design variables or to include them in the goal function and constraints.

In addition, a geometric pre-analysis and performance post-analysis, are implemented using Excel. The geometry pre-analysis is used to parametrize the design variables which are the pitch, β , thickness-ratio, t/c , and chord-to-radius ratio, c/R , distribution along the blade. The current effort uses a Bezier spline to achieve smooth distribution of the geometry, 5 parameters for each distribution. Thus, the design problem contains total of 15 design variables. All airfoils are based on the Mejzlik 18×6 cross sections and the propeller radius is $R=0.23$ m.

To ensure the structural properties of the optimized designs, two geometric constraints are to be satisfied. First the blade thickness distribution should not be lower than the original Mejzlik 18×6. Thin blade might be “soft” or exposed to high stresses, which might cause unacceptable aeroelastic behavior, and high deflections. In addition, the root chord should not be larger than the Mejzlik 18×6’s root chord. This might cause a very thick hub which increases the propeller weight.

To overcome these issues, the design procedure incorporated two geometric constraints over the thickness distribution and root chord. The first constraint limits the thickness distribution and the second the root chord. The thickness distribution, t (not t/c) has to be higher or equal to the Mejzlik 18×6 up to $r/R=0.90$, with a tolerance of 0.1 mm. The blade tip ($0.9 < r/R < 1$) was freed from this constraint – the impact over the design was high and it seems the importance of this constraint, at the very tip of the blade, is less important. The root chord is limited to $c/R < 0.15$. Note that the Mejzlik 18×6’s $c/R = 0.14$ at the root (Fig. 2), thus small increase of the root chord is allowed.

The performance post-analysis excel module is used to find the propeller-motor matching speed. Using the performance calculation, for given propeller geometry, several rotational speeds are calculated. To find the correct rotational speed, which the propeller produces the required thrust, $T=2.8$ kgf, a linear interpolation is used. Then, using the electric model, Eqs. (2)-(4), the electric power, P_e , is found.

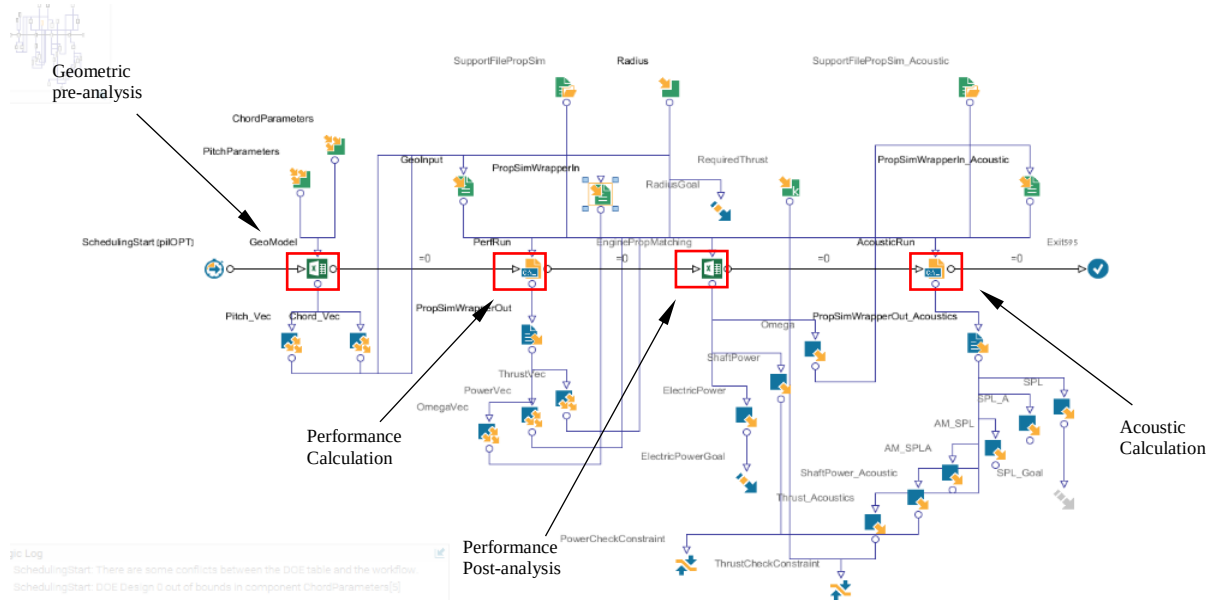


Fig. 12 modeFRONTIER design framework screen capture

The optimization was conducted 3 times for 2, 3, and 4 blades configurations. Pareto frontiers which resulted from the optimization are presented in Fig. 13. The optimization method and additional details concerning the solution procedure are available in Ref. [21].

In Fig. 13 propellers based on the Mejzlik 18×6 blades are marked by red circles. The influence of number-of-blades is prominent – increased number-of-blades causes both tonal-OASPL decrease and P_e increase.

From the Pareto frontiers, 4 propeller configurations were chosen – these are marked with arrows in Fig. 13 and include:

- 2 blades, minimum P_e (2B min P_e)
- 2 blades, minimum SPL (2B min SPL)
- 3 blades, minimum P_e (3B min P_e)
- 4 blades, minimum P_e (4B min P_e)

The propeller characteristics are depicted in Table 1, and their blade geometric parameters in Fig. 14. The clear difference is the rotational speed. This appears both as the mechanism of reducing the OASPL for the 2 blades propeller and for achieving the proper thrust for the 3 and 4 bladed propeller. To reduce the rotational speed, thus achieving min SPL for the 2 bladed propeller, the pitch was increased and the chord slightly increased.

For the 3 and 4 blades, the rotational speed had to decrease to achieve the required $T=2.8$ kgf. The chord cannot decrease due to the geometric constraint, thus the chord remained similar and thickness remains above the Mejzlik 18×6 blade. To maintain high enough rotational speed, the pitch decreased for the 3 and 4 bladed propellers, thus the electric efficiency and FM remain relatively high.

While the 3 bladed propeller exhibits high FM and low η_e , the 4 bladed exhibits low FM and high η_e . Generally, all tradeoff in such complex design case, is beyond simple intuition and it is a result of handling with all constraints while striving to minimize all design goals. This proves the advantage of such MDO (multidisciplinary design optimization) framework, which takes contradicting requirements and find the best compromise.

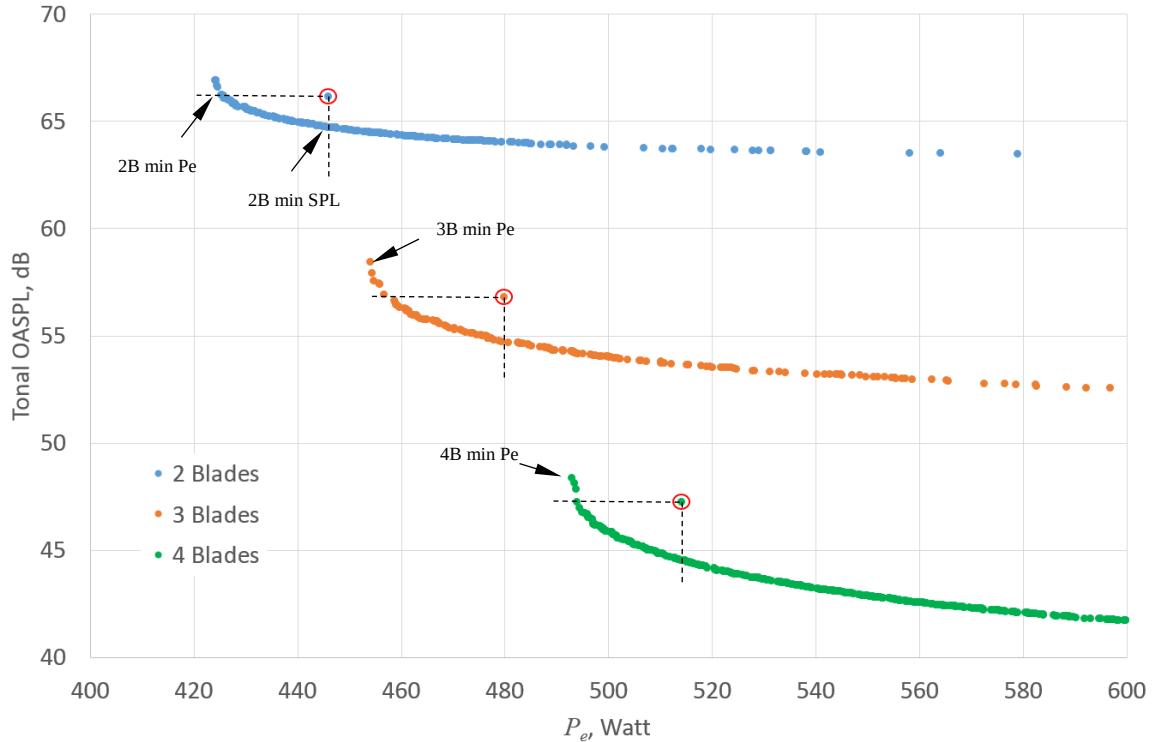


Fig. 13 Pareto frontiers for the optimized results
Red circles mark the results for propeller based on Mejzlik 18×6 blades

Table 1 optimized propeller characteristics

	2-Blades Mejzlik 18x6	2-Blades min. P_e	2-Blades min. SPL	3-Blades min. P_e	4-Blades min. P_e
Electric Power, P_e , W	445	425	445	455	495
Shaft Power, P_{shaft} , W	340	335	345	350	375
Ω , rpm	5,100	5,200	4,600	4,700	4,500
FM	0.68	0.69	0.67	0.66	0.62
Electric efficiency, η_e	0.77	0.79	0.77	0.67	0.76
Tonal SPL, dB	66.1	66.1	64.7	56.9	48.3

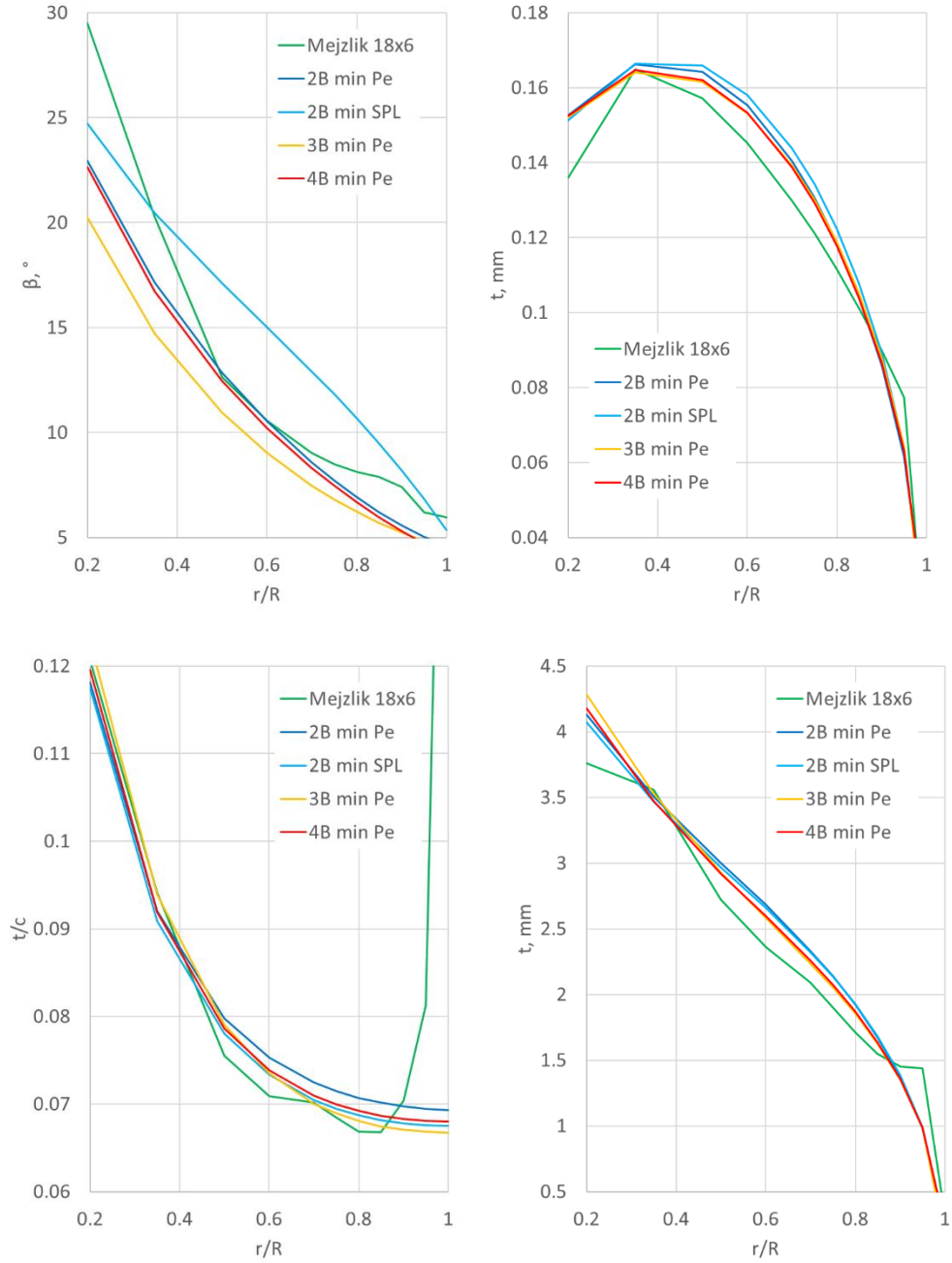


Fig. 14 Optimized blade geometries

VII. Test of Optimized Designs

The four optimized designs were fabricated and then tested on the test rig for both their performance and acoustic characteristics. The performance test was done on the same test rig which measured the thrust, shaft power, electric power and rotational speed. Fig. 17 shows the optimized propellers and their installation in the anechoic chamber.

The thrust as function of motor speed is presented in Fig. 15. The curves are quadratic regression which fits the theory of hover propellers – their thrust is proportional to the square of the rotational speed. From the right chart in Fig. 15 one can find the propeller rotational speed at the desired thrust, $T=2.8$ kgf.

P_e versus $T^{1.5}$ results are presented in Fig. 16 as. In this method, the theoretical linear relation between $T^{1.5}$ and required power is shown. Simple linear regression is used to define the propeller performance curve. This enable an accurate estimation of the required power for achieving $T=2.8$ kgf from the right chart (zoomed) of Fig. 16.

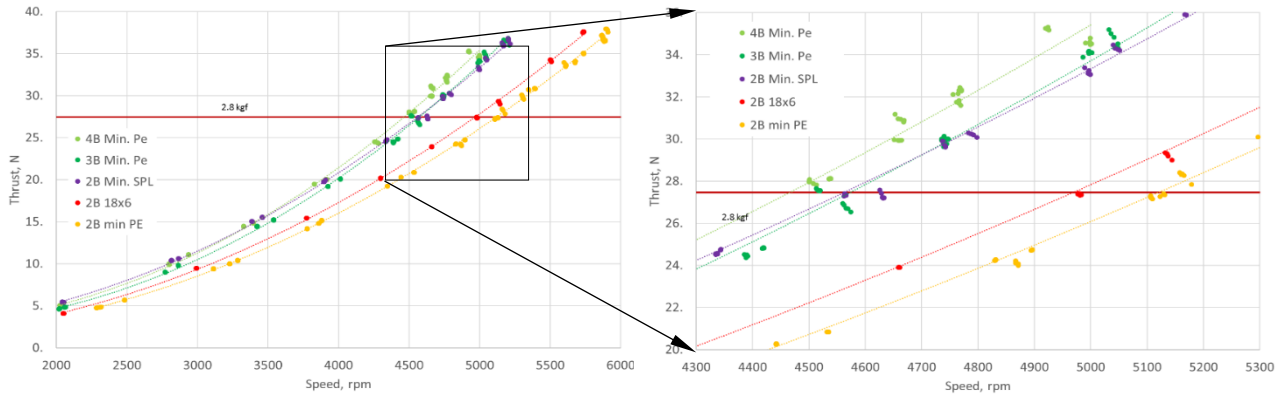


Fig. 15 Thrust as function of motor speed
Right chart is zoomed

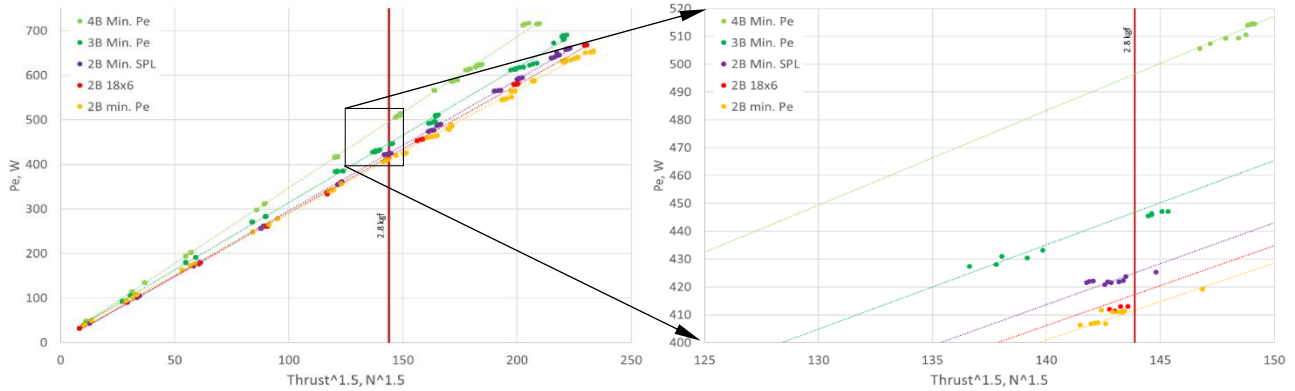


Fig. 16 Required electric power vs. thrust^{1.5}
Right chart is zoomed

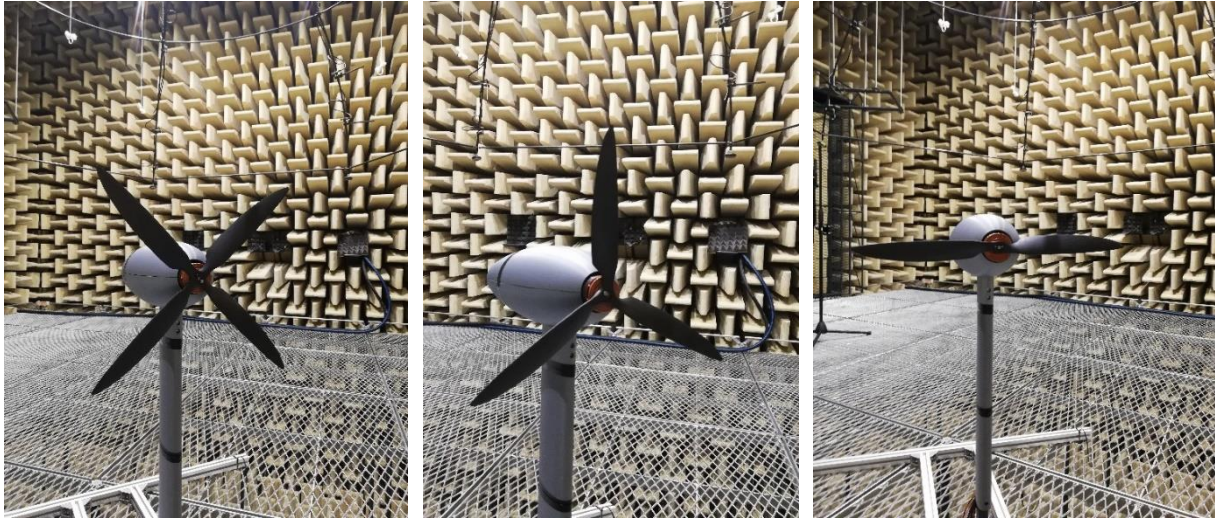
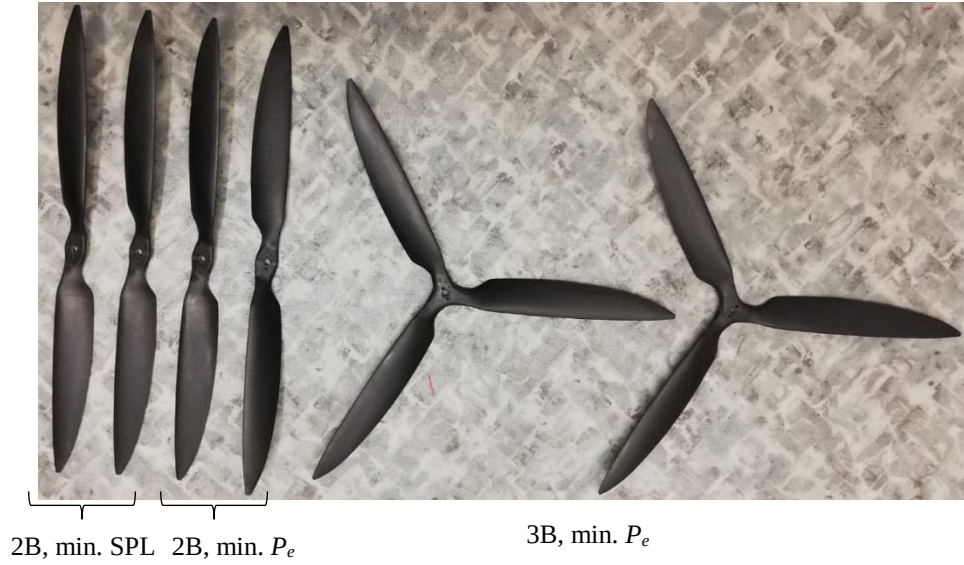


Fig. 17 Optimized propellers

Using the above results, the performance at the design conditions, $T=2.8$ kgf can be found and compared to the design estimations, thus validate the design procedure (performance-wise first). Fig. 18 presents the same zoomed view of the thrust and electric power in Fig. 15 and Fig. 16. The added solid lines represent the analysis. The design condition of $T=2.8$ kgf is marked on the charts, thus the design performance can be found and are shown in Table 2. It seems that the required electric power is over-predicted, for all 2-blades configurations by $15\text{ W} \div 25\text{ W}$ ($4\% \div 6\%$). For the 3 and 4 bladed propellers, the results are more accurate and the analysis is over-predicted by no more than 15 W (3%). Considering the rotational speed, the analysis is over-predicting by up to 150 rpm (3%). These errors seem reasonable, considering the complex relationship between all parameters.

More interesting is the design trends. The last 2 rows of Table 2 use the Mejzlik 18×6 as a reference, and the electric power difference, together with rotational speed difference are shown. This is done twice, once for the test results and then for the analysis (in parentheses). All electric power and rotational speed differences are predicted accurately. Thus, validates the design procedure process.

Table 2 optimized propeller performance test vs analysis (in parenthesis with difference percentage)

	2-Blades Mejzlik 18x6	2-Blades min. P_e	2-Blades min.SPL	3-Blades min. P_e	4-Blades min. P_e
P_e , W	415 (435, 5%)	410 (425, 4%)	425 (450, 6%)	445 (460, 3%)	495 (495, 0%)
P_{shaft} , W	325 (335, 3%)	320 (330, 3%)	320 (340, 6%)	330 (345, 5%)	360 (365, 1%)
Ω , rpm	5,000 (5,150, 3%)	5,100 (5,200, 2%)	4,600 (4,650, 1%)	4,600 (4,700, 2%)	4,450 (4,500, 1%)
ΔP_e , W (18x6 is ref.)	0	-5 (-10)	10 (15)	20 (25)	80 (60)
$\Delta \Omega$, rpm (18x6 is ref.)	0	100 (50)	-400 (-500)	-400 (-450)	-550 (-650)

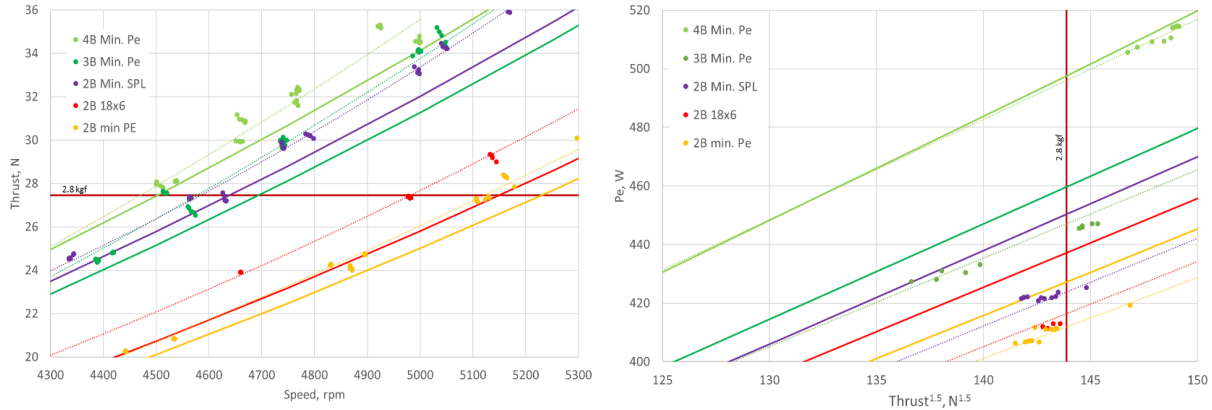


Fig. 18 Thrust vs. rotational speed and P_e vs. $T^{1.5}$
Dots – test, Dash line – test regression, Solid line - analysis

All optimized and reference propellers went into anechoic chamber acoustic test and their tonal SPL were measured. Two test series were conducted: thrust of 2.8 kgf, and speed of 5,000 rpm. These two different condition will shed light on the noise accuracy of the used analysis and the design trends as predicted by the optimization scheme.

For thrust of 2.8 kgf, the results are presented in Fig. 19 and for 5,000 rpm results are presented in Fig. 20. Both chart presents 1st harmonic and tonal-OASPL difference between the optimized propeller and the reference Mejzlik 18x6. The results are presented as function of the microphone θ angle. Note the data is taken from microphones which are located on different distances from the propeller. Thus, no conclusion can be deducted from the OASPL or SPL vs. θ dependency from the present figures.

The main purpose of this comparison is to compare the trends of test and analysis. The delta is calculated by subtracting the various propellers data from the reference Mejzlik 18x6. This shows the design trend as found by the measurements and by analysis. For low azimuth angles, θ , which exhibits high SPL, the comparison is good. Difference is less than 1 dB which exhibits good validation of the analysis. Moreover, the design trends are correct, thus the optimization procedure is also validated.

The minimum P_e propeller was designed to maintain the same noise level and to reduce the required power as compared to the Mejzlik 18x6 propeller. The former is kept in a very good way – the difference between the minimum P_e and the reference Mejzlik 18x6 is almost zero.

The minimum SPL propeller reduced noise level (compared to the 18x6) is reproduced in a good manner by the analysis.

The 3-bladed min P_e propeller's SPL is improved by about 10 dB – this is again predicted correctly by the analysis compared to the anechoic chamber test. Also the 4 bladed, exhibits good results in the limited available data (forward azimuth for the “5,000 rpm case”).

As θ increases, the microphone distance increase, thus the SPL decrease. This brings the SPL level within the 60÷50 dB range, for which the test result analysis is questionable. This is the reason that high θ , for high number-of-blade propellers is not presented.

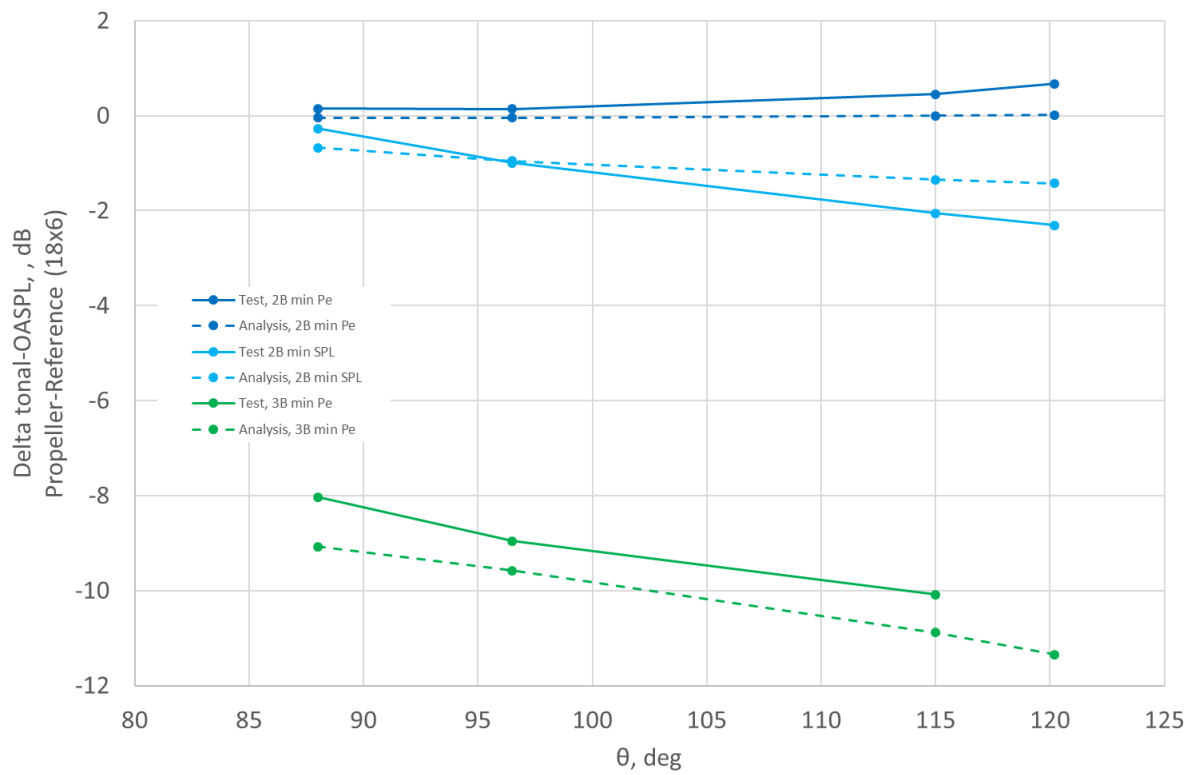
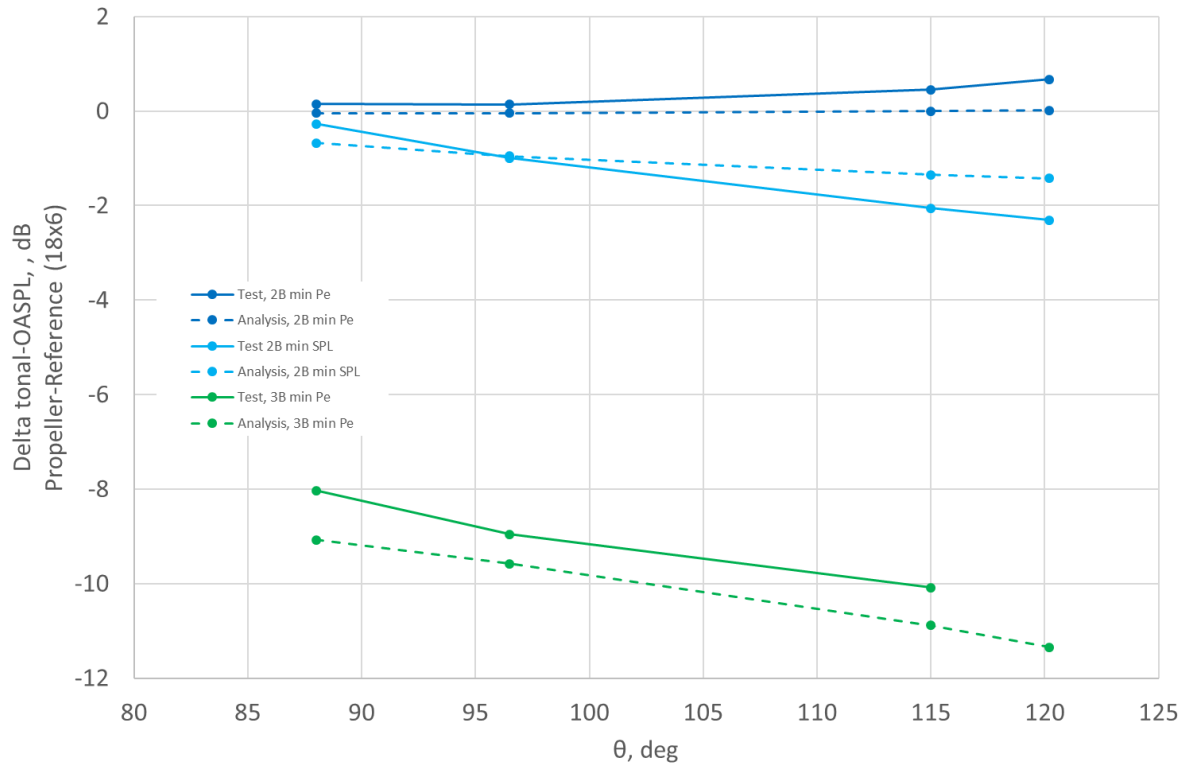


Fig. 19, Delta of 1st harmonic SPL and OASPL comparison vs. θ , Thrust=2.8 kgf

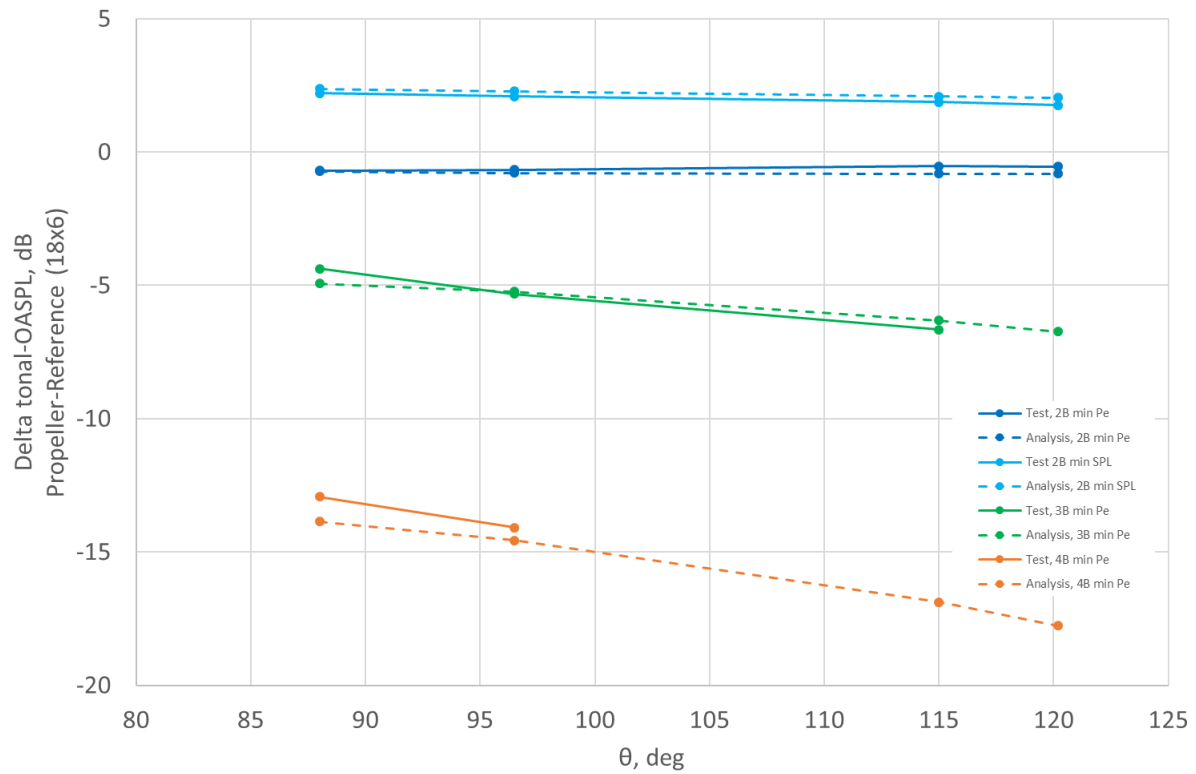
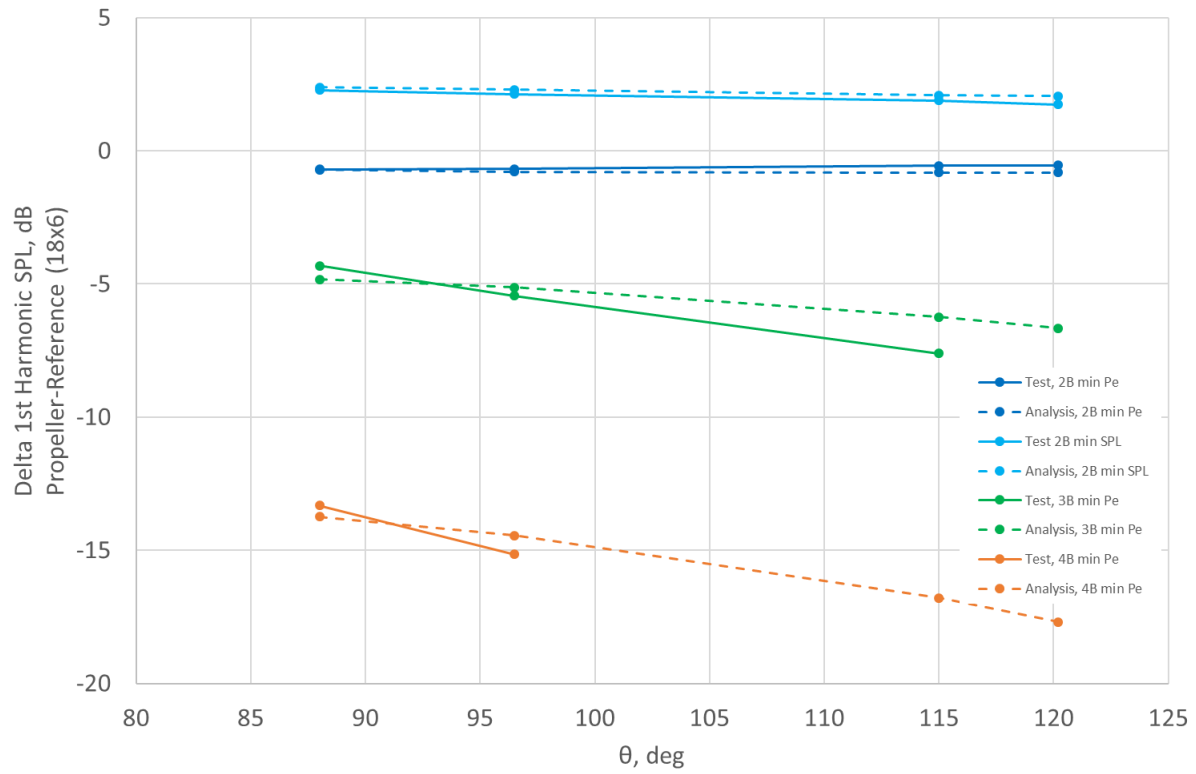


Fig. 20, Delta of 1st harmonic SPL and OASPL comparison vs. θ , $\Omega=5,000\text{rpm}$

As stated above, capability of the current measurement to detect the tonal SPL, for SPL lower than ~50 dB is poor. This is due to the influence of all noise components, except the tonal: background, motor, and broadband. Each of these contributions is hard to quantify, but the ~50 dB threshold is very distinct along the entire results.

To demonstrate this claim, measurements of microphone at $\theta=88^\circ$ of 3 different propellers: 2, 3, and 4 blades minimum Pe optimized propellers are given in Fig. 21. It is noticeable that all results predict the first harmonics with good accuracy. As tonal harmonics decrease below 50 dB, the high harmonic measurement becomes almost constant. Hence, other noise components become higher than the tonal. As the propeller is louder, higher number of harmonic is detected. Solid first 3 harmonics for the 2 blades, 2 harmonics for the 3 blades and only the first for the 4 blades. This also apparent for the OASPL which is accurately predicted for the 2 blades and deteriorates as number-of-blades increases.

The same phenomenon is replicated concerning the azimuth of the microphone via a comparison between the rear microphones ($\theta \geq 115^\circ$) and the front ($\theta \leq 115^\circ$). As azimuth angle increases, the microphone absolute distance from the propeller increases, due to the microphone installation in the chamber. Thus, the rear microphones detect lower noise level, which brings it closer to the ~50 dB threshold. Comparison of 3 different azimuth angles for the 2 blades minimum Pe propeller is given in Fig. 22. These charts again exhibit the ~50 dB threshold of the current measurements. While the most forward microphone located at $\theta=88^\circ$ detect solid 3 harmonics, the most aft microphone ($\theta=150^\circ$) hardly detects the first. This 50 dB threshold also causes the 4-bladed propeller tonal-noise to be hard to measured. This is especially true for the rear microphones of the 4-bladed propellers which makes the 4 bladed propeller test results quite questionable.

The good match between the test trends and analysis trends should be considered in the scope of tonal vs. total OASPL. As mentioned above, the total OASPL contains the background, motor, and broadband noise on top of the tonal OASPL. Fig. 23 shows the OASPL of the 2, 3, and 4 blades optimized minimum Pe propellers. The results are given for the tonal (both test and analysis) and the total OASPL.

For the 2 bladed propellers, there is almost a constant difference of 2dB between the total and tonal OASPL, for the azimuth of $\theta \leq 120^\circ$. For the $\theta=150^\circ$ azimuth, the difference increases dramatically to ~10 dB. The 3 blades and 4 blades exhibit even higher differences which can reach 15 dB and more.

An interesting common trend, is the minimal value of the total OASPL which reaches 75 dB. This might imply again on some measurement capability threshold, although 75 dB is quite high.

Another issue is the importance of the tonal versus other noise sources. Importance meaning not just the OASPL, but also the perception of noise by the human hear. This is more of a psycho-acoustic issue which is not dealt in the current scope.

Fig. 24 presents the 1st harmonic test results via the propeller's directivity map. Directivity is another way of presenting the data that allows one to grasp the sound radiation pattern from the propeller, at which positions will a propeller sound louder or quieter for a fixed observer distance. Here, results from all microphones that were measured, are presented. To compare the 5 test cases all microphones were normalized to a nominal radius of 1 meter. Since the microphones were placed on a hemisphere above the propeller, several microphones will have similar angular locations and will allow for a symmetry check. The results reveal there are no asymmetrical anomalies, and with the exception of the 3 and 4 bladed propellers at angles close to 90° , whose amplitudes are the smallest, results are consistent. Similar to the above results, lack of difference between the cases for positions behind the propeller is due to the low SPL

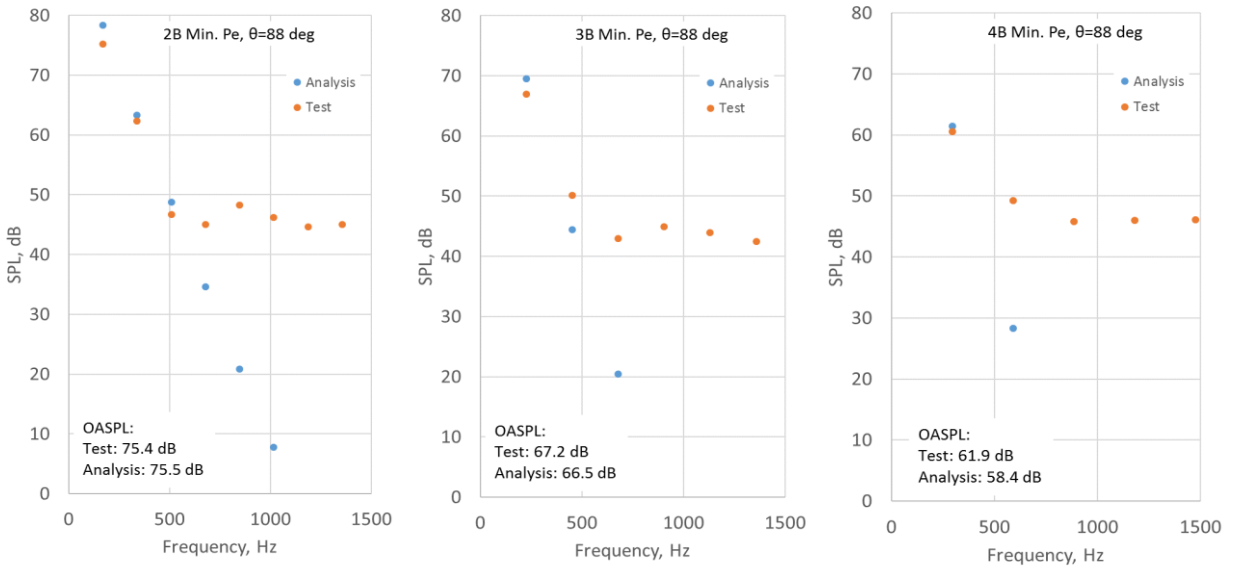


Fig. 21, $\theta=88^\circ$ measurements, 2, 3, and 4 blades minPe, optimized propellers, Thrust=2.8 kgf

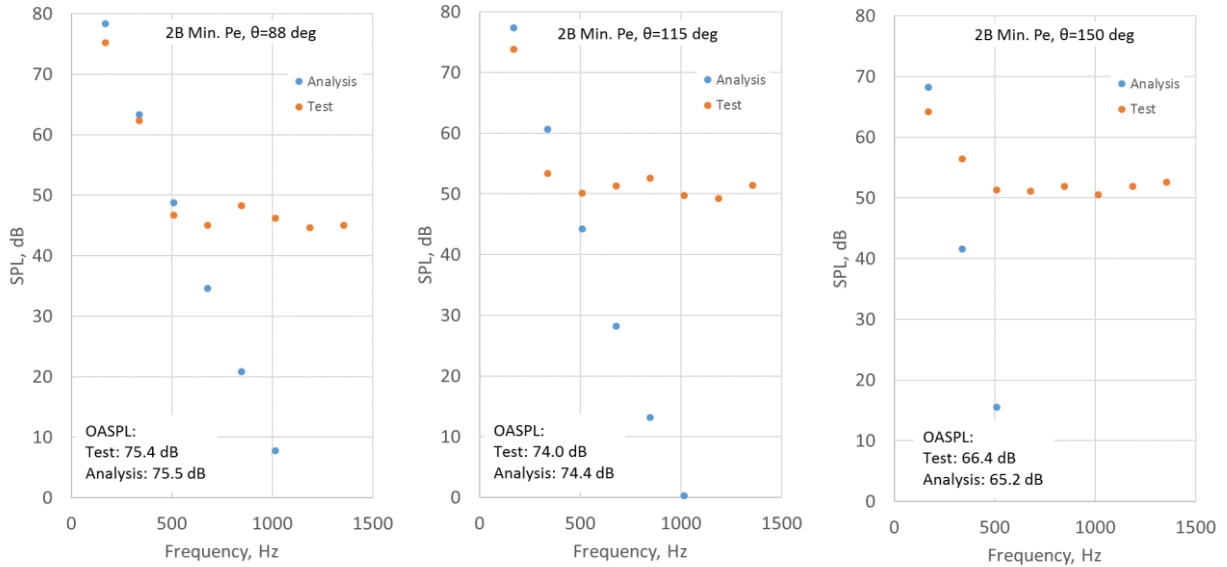


Fig. 22, 2 blades, minimum P_e propeller acoustic measurements, Thrust=2.8 kgf

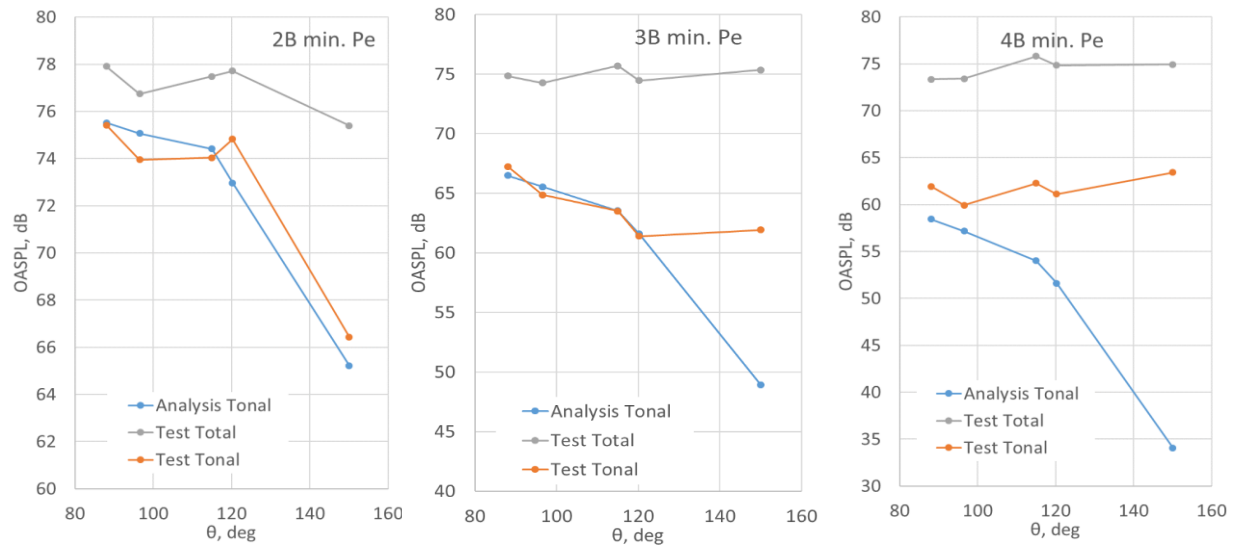


Fig. 23, Tonal vs. total OASPL results

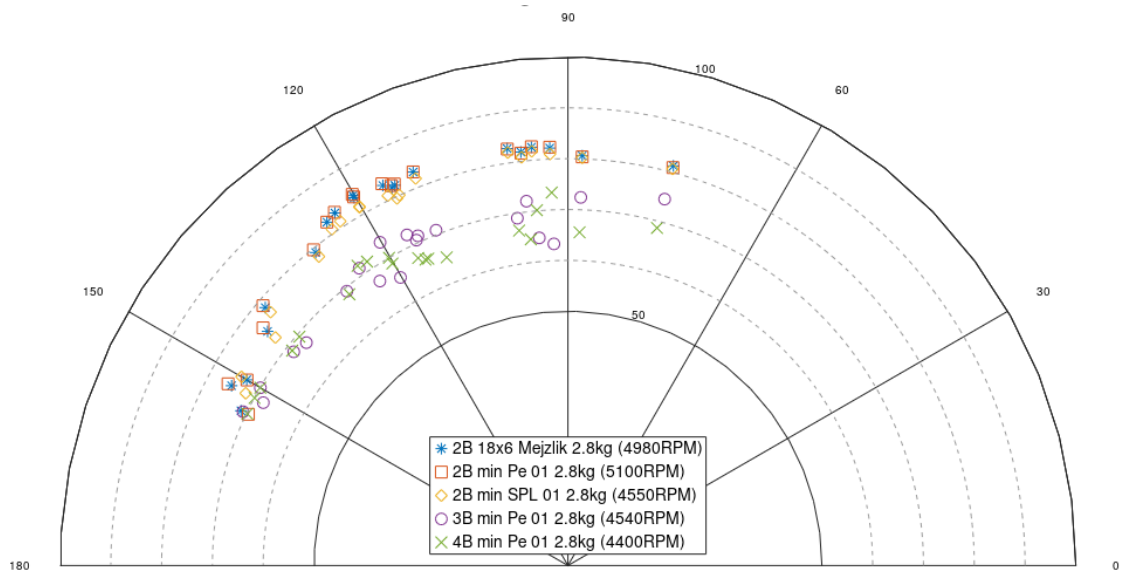


Fig. 24 1st Harmonic Propeller Directivity. All SPLs have been normalized to a reference distance of 1 meter.

VIII. Conclusion

In this paper a comprehensive and methodic design process for hover-propeller is described. The design process requires a detailed specification. In the current case, an existing propulsion system with of-the-shelf propeller is considered as a reference for the system specification. For this specification, an optimization design framework was developed which allows the optimized design of various propeller configurations.

In the basis of the design process are 3 analytic models: blade-element model for the propeller performance, electric model for the propulsion system characteristics, and acoustic model which analyzes the propeller tonal sound-pressure-level. Each of these models was previously validated versus various results in the literature. Still, in the current case, dedicated tests were conducted to improve the validation of the design analyses.

After the validation process, the analyses were incorporated in a design framework based on modeFRONTIER software and a multidisciplinary-design-optimization environment was substantiated. This environment includes, beside the analyses, various definitions of design variables, constraints, and design goals. Hence a multi-objective optimization problem is defined.

The design framework was run 3 times for designing 2, 3, and 4 bladed propellers. Each run resulted with a Pareto frontier which exhibits the tradeoff between the propulsion-system performance and its acoustic signature. From these tradeoffs, optimized propeller configurations were chosen. These are then fabricated and tested. The test results for both performance and acoustics is then compared with the design trends, thus the design process is validated.

In the current effort 4 propeller were fabricated. Two of them are 2 bladed, minimal electric power and minimal acoustic signature. In addition, 3 bladed and 4 bladed propellers for minimum electric power were chosen. The four propellers exhibited accurate performance and acoustic trends, thus validated the design process.

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